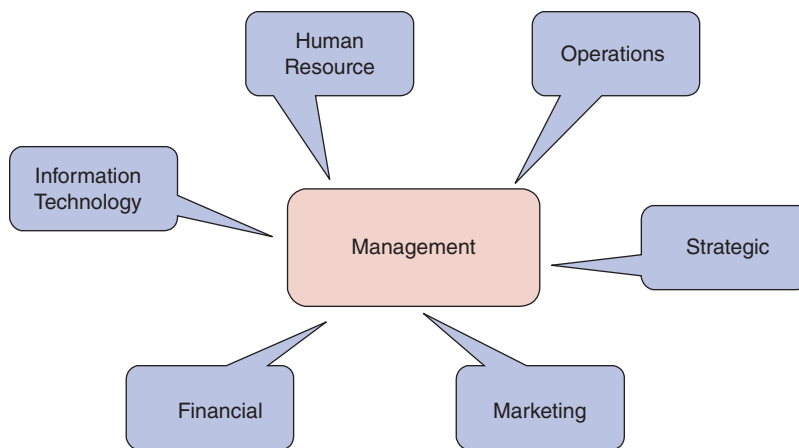


Conceptual Foundations of Accounting Information Systems

PART

I



CHAPTER 1

Accounting Information
Systems: An Overview

CHAPTER 2

Overview of Transaction
Processing and Enterprise
Resource Planning Systems

CHAPTER 3

Systems Documentation
Techniques

Accounting Information Systems: An Overview

LEARNING OBJECTIVES

- 1 Distinguish data from information, discuss the characteristics of useful information, and explain how to determine the value of information.
- 2 Explain the decisions an organization makes, the information needed to make them, and the major business processes present in most companies.
- 3 Explain how an accounting information system (AIS) adds value to an organization, how it affects and is affected by corporate strategy, and its role in a value chain.

INTEGRATIVE CASE

S&S

After working for years as a regional manager for a retail organization, Sundip Patel opened his own business with Silvia Gonzalez, one of his district managers, as his partner. They formed S&S to sell appliances and consumer electronics and pursued a “clicks and bricks” strategy by renting a building in a busy part of town and adding an electronic storefront.

Sundip and Silvia invested enough money to see them through the first six months. They will hire 15 employees within the next two weeks—three to stock the shelves, four sales representatives, six checkout clerks, and two to develop and maintain the electronic storefront.

They will host S&S’s grand opening in five weeks. To meet that deadline, they have to address the following important issues:

1. What decisions do they need to make to be successful and profitable? For example:
 - a. How should they price products to be competitive yet earn a profit?
 - b. Should they extend credit, and, if so, on what terms? How can they accurately track what customers owe and pay?
 - c. How should they hire, train, and supervise employees? What compensation and benefits package should they offer? How should they process payroll?
 - d. How can they track cash inflows and outflows to avoid a cash squeeze?
 - e. What is the appropriate product mix? What inventory quantities should they carry, given their limited showroom space?



Ndul/123RF

2. What information do they need to make those decisions?
 - a. What information do the external entities they interact with need?
 - b. What information do management and other employees need?
 - c. How can they gather, store, and disseminate that information?
3. What business processes are needed, and how should they be carried out?
4. What functionality should be provided on the website?

Although Silvia and Sundip could use an educated guess or “gut feeling” to make these decisions, they know they can make better decisions if they obtain additional information. A well-designed AIS can solve these issues and provide the information they need to make any remaining decisions.

Introduction

We begin this chapter by explaining important terms and discussing the information organizations need and the business processes used to produce that information. We continue with an exploration of what an accounting information system (AIS) is, how an AIS adds value to an organization, how an AIS and corporate strategy affect each other, and the role of the AIS in the value chain.

A **system** is a set of detailed methods, procedures, and routines that carry out specific activities, perform a duty, achieve goals or objectives, or solve one or more problems. Most systems are composed of smaller subsystems that support the larger system. For example, a college of business is a system composed of various departments, each of which is a subsystem. Moreover, the college itself is a subsystem of the university.

Each subsystem is designed to achieve one or more organizational goals. Changes in subsystems cannot be made without considering the effect on other subsystems and on the system as a whole. **Goal conflict** occurs when a subsystem’s goals are inconsistent with the goals of another subsystem or with the system as a whole. **Goal congruence** occurs when a subsystem achieves its goals while contributing to the organization’s overall goal. The larger the organization and the more complicated the system, the more difficult it is to achieve goal congruence.

Data are facts that are collected, recorded, stored, and processed by an information system. Businesses need to collect several kinds of data such as the activities that take place, the resources affected by the activities, and the people who participate in the activity. For example, the business needs to collect data about a sale (date, total amount), the resource sold (good or service, quantity sold, unit price), and the people who participated (customer, salesperson).

Information is data that have been organized and processed to provide meaning and context that can improve the decision-making process. As a rule, users make better decisions as the quantity and quality of information increase. Table 1-1 presents 14 characteristics that make information useful and meaningful.

system - (1) Two or more interrelated components that interact to achieve a goal, often composed of subsystems that support the larger system.
 (2) The equipment and programs that comprise a complete computer installation.
 (3) The detailed methods, procedures, and routines that carry out specific activities, perform a duty, achieve goals or objectives, or solve one or more problems.

goal conflict - When a subsystem’s goals are inconsistent with the goals of another subsystem or the system as a whole.

goal congruence - When a subsystem achieves its goals while contributing to the organization’s overall goal.

data - Facts that are collected, recorded, stored, and processed by an information system.

information - Data that have been organized and processed to provide meaning and improve decision making.

TABLE 1-1 Characteristics of Useful Information

Access restricted	Able to limit access to authorized parties
Accurate	Correct; free of error; accurately represents events and activities
Available	Available to users when needed; in a format that can be easily and quickly used
Complete	Does not omit aspects of events or activities; of enough breadth and depth
Concise	Clear, succinct; appropriate volume presented briefly but comprehensively
Consistent	Presented in same format over time
Current	Includes event and activity data up to the present date and time
Objective	Unbiased; unprejudiced; impartial
Relevant	Reduces uncertainty; improves decision making; applicable and helpful
Reputable	Perceived as true and credible due to highly regarded source or content
Timely	Provided in time for decision makers to make decisions
Understandable	Presented in a useful and intelligible format; easily comprehended and interpreted
Useable	Easy to use for different tasks; human and machine readable
Verifiable	Same information produced by two independent, knowledgeable people

machine-readable - Data in a format that can be read and processed by a computer.

Data are most useful when presented in a **machine-readable** format that can be read and processed by a computer. This processing may involve data collection, recording, storage, updating, and data dissemination. For example, public companies are now required to code their financial statements using XBRL (eXtensible Business Reporting Language)—a programming language designed specifically to facilitate the communication of financial and other business information. Without XBRL, electronic documents are digital versions of paper reports. Humans can read the data, but computers cannot automatically process the data until a person manually enters it in the appropriate format. XBRL changes that by encoding information about what a particular data item means so that other computer programs can understand what to do with it. XBRL is discussed more fully in **Chapter 16**.

Using machine-readable formats like XBRL improves many of the other 14 characteristics that make information useful. For example, XBRL improves:

- Reliability by reducing human error and using standard taxonomies.
- Relevance by assigning relevant meaning to data so it can be compared to similar data from other organizations.
- Accessibility by enabling the automatic importing of data into decision models and other computer systems.
- Understandability and usability by making the data readable to both humans and computers.
- Timeliness by reducing the time needed to import, produce, and distribute information.

Machine-readable data also facilitate the use of technologies such as artificial intelligence and data analytics, which are discussed later in the chapter.

However, there are limits to the amount of information the human mind can absorb and process. **Information overload** occurs when those limits are passed, resulting in a decline in decision-making quality and an increase in the cost of providing that information. Information system designers use **information technology (IT)** to help decision makers more effectively filter and condense information.

The **value of information** is the benefit produced by the information minus the cost of producing it. Benefits of information include reduced uncertainty, improved decisions, and improved ability to plan and schedule activities. The costs include the time and resources spent to produce and distribute the information. Information costs and benefits can be difficult to quantify, and it is difficult to determine the value of information before it has been

information overload - Exceeding the amount of information a human mind can absorb and process, resulting in a decline in decision-making quality and an increase in the cost of providing information.

information technology (IT) - The computers and other electronic devices used to store, retrieve, transmit, and manipulate data.

value of information - The benefit provided by information minus the cost of producing it.

produced and utilized. Nevertheless, the expected value of information should be calculated as effectively as possible so that the costs of producing the information do not exceed its benefits.

To illustrate the value of information, consider the case of 7-Eleven. When a Japanese company licensed the very successful 7-Eleven name from Southland Corporation, it invested heavily in IT. However, the U.S. stores did not. Each 7-Eleven store in Japan was given a computer that:

- Keeps track of the 3,000 items sold in each store and determines what products are moving, at what time of day, and under what weather conditions.
- Keeps track of what customers buy, when they buy it, and when to purchase additional inventory to replace what was sold.
- Orders sandwiches and rice dishes from suppliers automatically. Orders are placed and filled three times a day so that stores always have fresh food. In addition, suppliers can access 7-Eleven sales data electronically so that they can forecast demand.
- Coordinates deliveries with suppliers. This reduces deliveries from 34 to 12 a day, resulting in less clerical receiving time.
- Prepares a color graphic display that indicates which store areas contribute the most to sales and profits.

Average daily sales of 7-Eleven Japan were 30% higher than those of its closest competitor and its operating margins were almost double. What happened to Southland and its 7-Eleven stores in the United States? Profits declined, and Southland eventually had to file for bankruptcy. 7-Eleven Japan came to the company's rescue and purchased 64% of Southland.

As shown in Figure 1-1, an **information system** is the combination of the people and the technologies in an organization that collect, record, store, and process data to produce the information needed to make informed decisions.

information system - The people and technologies in an organization that produce information. It is an organized way of collecting, processing, managing, and reporting information so that an organization can achieve its objectives and goals. Formal information systems have an explicit responsibility to produce information. An informal information system meets a need that is not satisfied by a formal channel and operates without formal designation of responsibility.

Information Needs and Business Processes

All organizations need information in order to make effective decisions. In addition, all organizations have certain business processes in which they are continuously engaged. A **business process** is a set of related, coordinated, and structured activities and tasks that are performed by a person, a computer, or a machine and that help accomplish a specific organizational goal.

To make effective decisions, organizations must decide what decisions they need to make, what information they need to make the decisions, and how to gather and process the data needed to produce the information. This data gathering and processing is often tied to the basic business processes in an organization. To illustrate the process of identifying information needs and business processes, let's return to our S&S case study.

business process - A set of related, coordinated, and structured activities and tasks, performed by a person, a computer, or a machine, that helps accomplish a specific organizational goal.

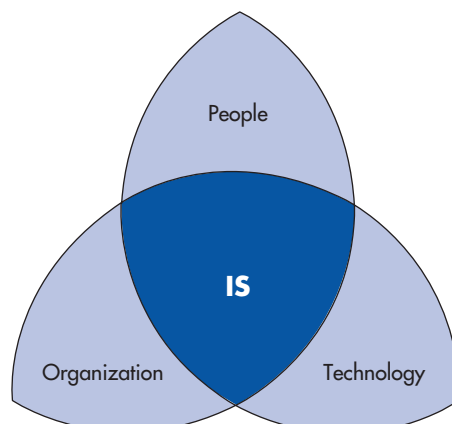


FIGURE 1-1
The Components of an Information System

INFORMATION NEEDS

Sundip and Silvia decide they must understand how S&S functions before they can identify the information they need to manage S&S effectively. Then they can determine the types of data and procedures they will need to collect and produce that information. They created Table 1-2 to summarize part of their analysis. It lists S&S's basic business processes, some key decisions that need to be made for each process, and information they need to make the decisions.

TABLE 1-2 Overview of S&S's Business Processes, Key Decisions, and Information Needs

Business Processes	Key Decisions	Information Needs
Acquire capital	How much Find investors or borrow funds If borrowing, how to obtain the best terms	Cash flow projections Pro forma financial statements Loan amortization schedule
Acquire building and equipment	Size of building Amount of equipment Rent or buy Location	Capacity needs Building and equipment prices Market study Average area household income, traffic patterns, ease of customer access
Hire and train employees	How to depreciate Experience requirements How to assess integrity and competence of applicants How to train employees	Tax tables and depreciation regulations Job descriptions Applicant job history and skills Skills needed; training objectives
Acquire inventory	What models to carry Optimal level of inventory to carry How much to purchase How to manage inventory (store, control, etc.)	Market analyses Sales and inventory turnover forecasts Sales projections; inventory status reports Inventory best practices and control procedures
Advertising and marketing	Which vendors; best quality; prices Which media	Vendor prices; performance reports Media alternatives; market coverage; cost analyses
Sell merchandise	Content What the optimal price is for each product How to customize products for customers How to deliver products to each customer Offer in-house credit?; determine credit limits Which credit cards to accept	Marketing best practices Product costs; desired margins Customer needs and preferences Customer delivery preferences Customer credit status Credit card costs; company policies
Collect payments from customers	If offering credit, what terms How to effectively handle cash receipts	Customer account limits and status; accounts receivable aging report Banking options; cash receipts best practices
Pay employees	Amount to pay Deductions and withholdings	Pay rate analysis and comparisons; sales (for commissions); time worked (hourly employees) W-4 forms
Pay taxes	Process payroll in-house or use outside service Payroll tax requirements Sales tax requirements	Costs of external payroll service Government regulations; total wage expense Government regulations; total sales
Pay vendors	When and whom to pay How much to pay	Vendor invoices; accounts payable records Vendor invoices; payment terms

Silvia and Sundip realize the list is not exhaustive, but they are satisfied it provides a good overview of S&S. They also recognize that not all the information needs listed in the right-hand column will be produced internally by S&S. Information about payment terms for merchandise purchases, for example, will be provided by vendors. Thus, S&S must effectively integrate external data with internally generated data so they can use both types of information to run S&S.

S&S will interact with many external parties, such as customers, vendors, and governmental agencies, as well as with internal parties such as management and employees. To get a better handle on the more important interactions with these parties, they prepared Figure 1-2.

BUSINESS PROCESSES

Sundip decides to reorganize the business processes listed in Table 1-2 into groups of related transactions. A **transaction** is an agreement between two entities to exchange goods or services or any other event that can be measured in economic terms. Examples include selling goods to customers, buying inventory from suppliers, and paying employees. The process that begins with capturing transaction data and ends with informational output, such as the financial statements, is called **transaction processing**. Transaction processing is covered in more depth in **Chapter 2**.

Many business activities are pairs of events involved in a **give-get exchange**. Most organizations engage in a small number of give-get exchanges, but each type of exchange happens many times. For example, S&S will have thousands of sales to customers every year in exchange for cash. Likewise, S&S will continuously buy inventory from suppliers in exchange for cash.

These exchanges can be grouped into five major **business processes or transaction cycles**:

- The **revenue cycle**, where goods and services are sold for cash or a future promise to receive cash. This cycle is discussed in **Chapter 14**.
- The **expenditure cycle**, where companies purchase inventory for resale or raw materials to use in producing products in exchange for cash or a future promise to pay cash. This cycle is discussed in **Chapter 15**.

transaction - An agreement between two entities to exchange goods or services, such as selling inventory in exchange for cash; any other event that can be measured in economic terms by an organization.

transaction processing - Process of capturing transaction data, processing it, storing it for later use, and producing information output, such as a managerial report or a financial statement.

give-get exchange - Transactions that happen a great many times, such as giving up cash to get inventory from a supplier and giving employees a paycheck in exchange for their labor.

business processes or transaction cycles - The major give-get exchanges that occur frequently in most companies.

revenue cycle - The recurring set of business activities and data processing operations associated with selling goods and services in exchange for cash or a future promise to receive cash.

expenditure cycle - A recurring set of business activities and related data processing operations associated with purchasing inventory or raw materials in exchange for cash or a future promise to pay cash.

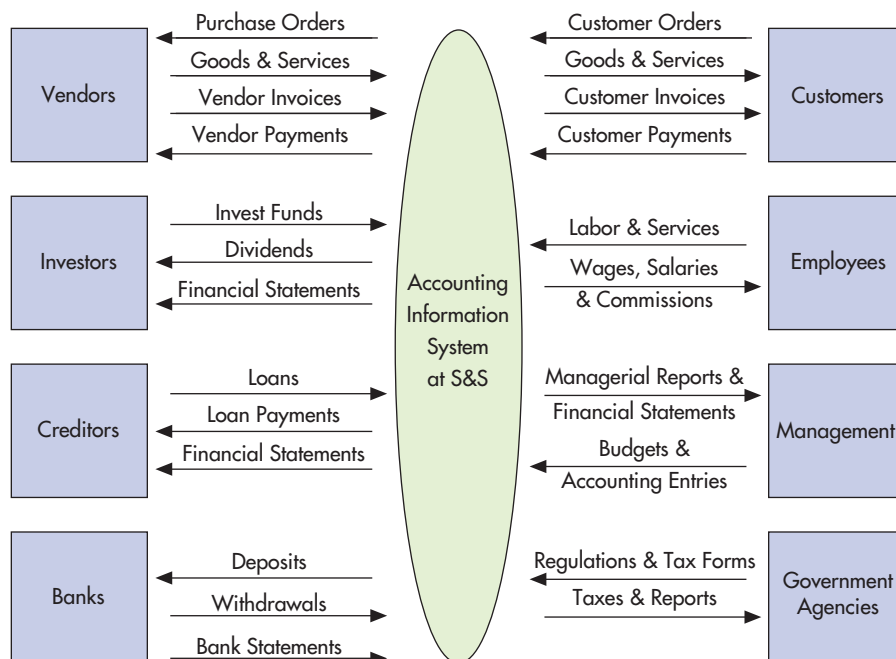


FIGURE 1-2
Interactions between
S&S and External and
Internal Parties

production cycle - The recurring set of business activities and related data processing operations associated with using labor, raw materials, and equipment to produce finished goods. Also called *conversion cycle*.

human resources/payroll cycle - Activities associated with hiring, training, compensating, evaluating, promoting, and terminating employees.

financing cycle - Activities associated with raising money by selling shares in the company to investors and borrowing money as well as paying dividends and interest.

- The **production cycle**, or conversion cycle, where raw materials are transformed into finished goods. This cycle is discussed in **Chapter 16**.
- The **human resources/payroll cycle**, where employees are hired, trained, compensated, evaluated, promoted, and terminated. This cycle is discussed in **Chapter 17**.
- The **financing cycle**, where companies sell shares in the company to investors and borrow money, and where investors are paid dividends and interest is paid on loans.

These cycles process a few related transactions repeatedly. For example, most revenue cycle transactions are either selling goods or services to customers or collecting cash for those sales. Figure 1-3 shows the main transaction cycles and the give-get exchange inherent in each cycle.

These basic give-get exchanges are supported by several other business activities. For example, S&S may need to answer a number of customer inquiries and check inventory levels before it can make a sale. Likewise, it may have to check customer credit before a credit sale is made. Accounts receivable will have to be increased each time a credit sale is made and decreased each time a customer payment is received. Table 1-3 lists the major activities in each transaction cycle.

Notice that the last activity listed in Table 1-3 for each transaction cycle is “Send appropriate information to the other cycles.” Figure 1-3 shows how these various transaction cycles

FIGURE 1-3
The AIS and Its
Subsystems

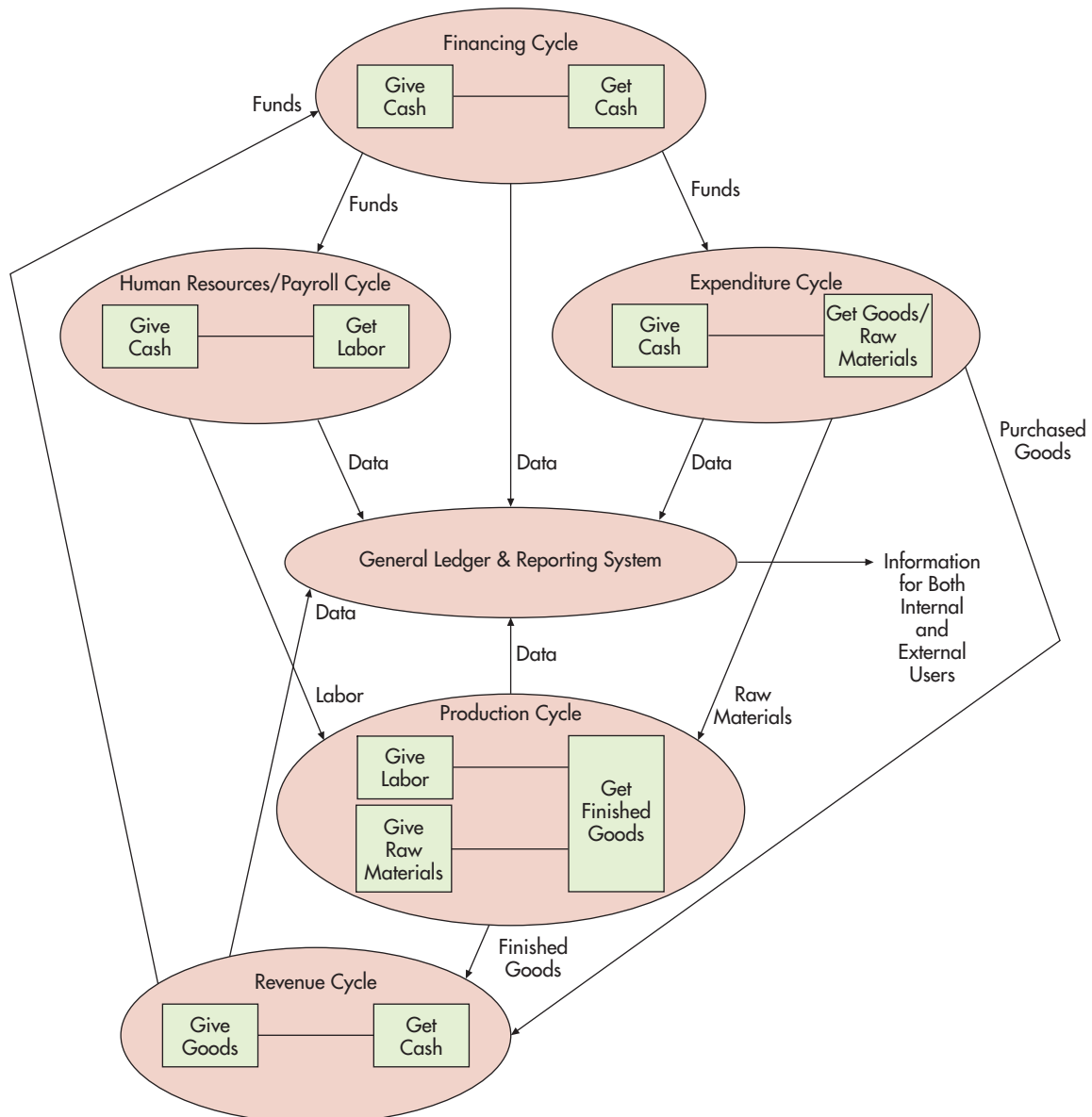


TABLE 1-3 Common Cycle Activities

Transaction Cycle	Major Activities in the Cycle
Revenue	Receive and answer customer inquiries Take customer orders and enter them into the AIS Approve credit sales Check inventory availability Initiate back orders for goods out of stock Pick and pack customer orders Ship goods to customers or perform services Bill customers for goods shipped or services performed Update (increase) sales and accounts receivable Receive customer payments and deposit them in the bank Update (reduce) accounts receivable Handle sales returns, discounts, allowances, and bad debts Prepare management reports Send appropriate information to the other cycles
Expenditure	Request goods and services be purchased Prepare, approve, and send purchase orders to vendors Receive goods and services and complete a receiving report Store goods Receive vendor invoices Update (increase) accounts payable Approve vendor invoices for payment Pay vendors for goods and services Update (reduce) accounts payable Handle purchase returns, discounts, and allowances Prepare management reports Send appropriate information to the other cycles
Human Resources/Payroll	Recruit, hire, and train new employees Evaluate employee performance and promote employees Discharge employees Update payroll records Collect and validate time, attendance, and commission data Prepare and disburse payroll Calculate and disburse taxes and benefit payments Prepare employee and management reports Send appropriate information to the other cycles
Production	Design products Forecast, plan, and schedule production Request raw materials for production Manufacture products Store finished products Accumulate costs for products manufactured Prepare management reports Send appropriate information to the other cycles
Financing	Forecast cash needs Sell stock/securities to investors Borrow money from lenders Pay dividends to investors and interest to lenders Retire debt Prepare management reports Send appropriate information to the other cycles

relate to one another and interface with the **general ledger and reporting system**, which is used to generate information for both management and external parties. The general ledger and reporting system is discussed in more depth in **Chapter 18**.

In many accounting software packages, the various transaction cycles are implemented as separate modules. Not every organization needs to implement every module. Retail stores

general ledger and reporting system - Information-processing operations involved in updating the general ledger and preparing reports for both management and external parties.

like S&S, for example, do not have a production cycle and would not implement that module. Moreover, some organizations have unique requirements. Financial institutions, for example, have demand deposit and installment-loan cycles that relate to transactions involving customer accounts and loans. In addition, the nature of a given transaction cycle differs across different types of organizations. For example, the expenditure cycle of a service company, such as a public accounting or a law firm, does not normally involve processing transactions related to the purchase, receipt, and payment for merchandise that will be resold to customers.

Each transaction cycle can include many different business processes or activities. Each business process can be relatively simple or quite complex.

After preparing Tables 1-2 and 1-3 and Figures 1-2 and 1-3, Sundip and Silvia believe they understand S&S well enough to begin shopping for an information system. Silvia recalled a previous employer that had several separate information systems because its software was not designed to accommodate the information needs of all managers. She also vividly recalled attending one meeting where she witnessed the negative effects of having multiple systems. The head of marketing had one report on year-to-date sales by product, the production manager had a different report that contained different sales figures, and the controller's report, which was produced by the general ledger system, had yet a third version of year-to-date sales. Over an hour was wasted trying to reconcile those different reports! Silvia vowed to ensure that S&S did not ever find itself in such a mess. She would make sure that any system selected would have the capability to integrate both financial and nonfinancial data about S&S's various business processes so that everyone could pull information from the same system.

Accounting Information Systems

accounting information system (AIS) - A system that collects, records, stores, and processes data to produce information for decision makers. It includes people, procedures and instructions, data, software, information technology infrastructure, and internal controls and security measures.

accounting - The systematic and comprehensive recording of an organization's financial transactions, including summarizing, analyzing, and reporting these transactions to all users.

It has often been said that accounting is the language of business. If that is the case, then an **accounting information system (AIS)** is the intelligence—the information-providing vehicle—of that language.

Accounting is the systematic and comprehensive recording of an organization's financial transactions. It also includes summarizing, analyzing, and reporting these transactions to management, owners/investors, oversight agencies, and tax collection entities. That means accounting is a data identification, collection, and storage process as well as an information development, measurement, and communication process. By definition, accounting is an information system, since an AIS collects, records, stores, and processes accounting and other data to produce information for decision makers. This is illustrated in Figure 1-4.

An AIS can be a paper-and-pencil manual system, a complex system using the latest in IT, or something in between. Regardless of the approach taken, the process is the same. The AIS must collect, enter, process, store, and report data and information. The paper and pencil or the computer hardware and software are merely the tools used to produce the information.

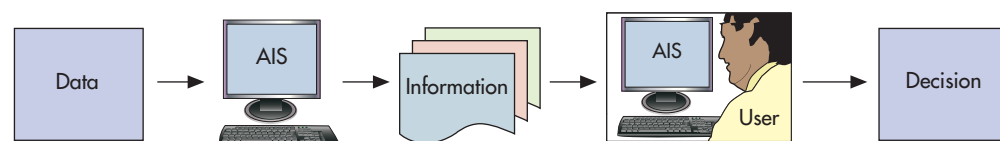
This text does not distinguish an AIS from other information systems. Instead, our viewpoint is that the AIS can and should be the organization's primary information system and that it provides users with the information they need to perform their jobs.

There are six components of an AIS:

1. The *people* who use the system.
2. The *procedures and instructions* used to collect, process, and store data.
3. The *data* about the organization and its business activities.
4. The *software* used to process the data.

FIGURE 1-4

An AIS Processes Data to Produce Information for Decision Makers



5. The *information technology infrastructure*, including the computers, peripheral devices, and network communications devices used in the AIS.
6. The *internal controls and security measures* that safeguard AIS data.

These six components enable an AIS to fulfill three important business functions:

1. Collect and store data about organizational activities, resources, and personnel. Organizations have a number of business processes, such as making a sale or purchasing raw materials, which are repeated frequently.
2. Transform data into information so management can plan, execute, control, and evaluate activities, resources, and personnel. Decision making is discussed in detail later in this chapter.
3. Provide adequate controls to safeguard the organization's assets and data. Control concepts are discussed in detail in **Chapters 8–13**.

Since accounting data comes from an AIS, knowledge and skills in AIS are critical to an accountant's career success. Interacting with an AIS is one of the most important activities accountants perform. Other important AIS-related activities include designing information systems and business process improvements, as discussed in **Chapters 22 to 24**. **Focus 1-1** explains a specialty to designate that certain CPAs (Certified Public Accountants) have an in-depth knowledge of AIS topics.

HOW AN AIS CAN ADD VALUE TO AN ORGANIZATION

A well-designed AIS can add value to an organization by:

1. **Improving the quality and reducing the costs of products or services.** For example, an AIS can monitor machinery so operators are notified immediately when performance falls outside acceptable quality limits. This helps maintain product quality, reduces waste, and lowers costs.

FOCUS 1-1 CITP—An IT Specialty Designation for CPAs

The American Institute of Certified Public Accountants (AICPA) offers several specialty designations for CPAs. The CITP (Certified Information Technology Professional) designation reflects the AICPA's recognition of the importance of IT and its interrelationship with accounting. A CITP possesses a broad range of business, managerial, and technological knowledge, making it possible for the CITP to understand how organizations use IT to achieve their business objectives. To obtain a CITP certification, a person must demonstrate a mastery of the following topics: information system management, business intelligence, fraud, risk assessment, internal control concepts, and how to test and evaluate an information system.

There are many reasons to earn the CITP certification:

- Because only CPAs can be CITPs, this certification further differentiates you from others in the marketplace.
- It affirms your value as an IT specialist and increases your value to your employer or clients.

- It is a great “calling card” for IT people who want to be leaders in industry, public practice, government, or academia.
- It opens the doors to new technology-related roles and opportunities.
- It provides an automatic membership in the IT Section, which allows you to meet, share best practices, network, and communicate with other CITPs. You can also receive CITP newsletters and other communications, attend CITP webinars, receive CITP member discounts, and access exclusive CITP resources and content on the CITP website. To qualify for the CITP designation, you must:
 - Be a CPA and a member of the AICPA
 - Pass the CITP exam and complete the CITP Credential Application
 - In the five years preceding your application, meet the minimum 1,000-hours of business experience in the CITP body of knowledge

Based on information from <http://www.aicpa.org>.

2. **Improving efficiency.** For example, timely information makes a just-in-time manufacturing approach possible, as it requires constant, accurate, up-to-date information about raw materials inventories and their locations.
3. **Sharing knowledge.** Sharing knowledge and expertise can improve operations and provide a competitive advantage. For example, CPA firms use their information systems to share best practices and to support communication between offices. Employees can search the corporate database to identify experts to provide assistance for a particular client; thus, a CPA firm's international expertise can be made available to any local client.
4. **Improving the efficiency and effectiveness of its supply chain.** For example, allowing customers to directly access inventory and sales order entry systems can reduce sales and marketing costs, thereby increasing customer retention rates.
5. **Improving the internal control structure.** An AIS with the proper internal control structure can help protect systems from fraud, errors, system failures, and disasters.
6. **Improving decision making.** Improved decision making is vitally important and is discussed below in more detail.

AN AIS CAN USE ARTIFICIAL INTELLIGENCE AND DATA ANALYTICS TO IMPROVE DECISION MAKING

Decision making is a complex, multistep activity that involves identifying a problem, collecting and interpreting data, evaluating ways to solve the problem, selecting a solution methodology, and determining and implementing the solution. An AIS can help in the decision-making process by providing the information to reduce uncertainty, providing feedback about the effectiveness of prior decisions, providing information in a timely manner, and identifying situations that require management action.

Artificial intelligence and data analytics tools can be used in each of these decision-making activities to help improve decision making.

Artificial intelligence (AI) uses computer systems to simulate human intelligence processes such as learning (acquiring information and rules for using it), reasoning (interpreting data and using its rules to arrive at conclusions), and self-improvement (learning from the information and past experiences to improve its rules).

The AI field draws from many disciplines, including computer science, information engineering, linguistics, mathematics, philosophy, and psychology. Some popular AI applications include expert systems, intelligent routing of delivery vehicles, machine vision (used in self-driving cars), and speech recognition. Many organizations, such as Amazon, Google, IBM, and Microsoft are imbedding AI features in their applications, tools, and search services.

Here are a few examples of the fields where AI is used:

- **Business.** AI-driven robots now perform many highly repetitive tasks, especially in manufacturing. AI algorithms determine how to better serve customers. Website chatbots provide immediate customer service.
- **Education.** AI software automatically grades student work, assesses student performance and progress, and provides additional support as needed.
- **Finance.** Robo-based stock-picking algorithms give advice on what stocks to buy and sell. Software executes most stock market trades. Personal finance applications use AI to advise their users and keep track of their finances.
- **Healthcare.** AI can make better and faster diagnoses than humans, thereby improving patient outcomes and reducing costs. For example, doctors can use IBM Watson to mine patient data, evaluate symptoms, access external databases, communicate a diagnosis (as well as how confident Watson is in the diagnosis), and receive and answer doctor questions.

Data analytics is the use of software and algorithms to discover, describe, interpret, communicate, and apply meaningful patterns in data to improve business performance. Data analytics tools draw from many disciplines, including computer programming, mathematics, operations research, and statistics. Companies have long analyzed their past performance. Data analytics is a more recent development and is designed to focus on the future and answer questions such as why something happened, what will happen next, and how performance can

artificial intelligence (AI) - The use of computer systems to simulate human intelligence processes such as learning, reasoning, and self-improvement.

data analytics - Use of software and algorithms to find and solve problems and improve business performance.

be improved. Analytics tools are most efficient when there are abundant computational resources and there is a large quantity of data from multiple internal or external databases.

An essential part of most analytic tools is a **data dashboard** that displays important data points, metrics, and key performance indicators in the form of line or bar charts, tables, or gauges. A dashboard provides a central location for businesses to monitor performance. The dashboard is connected to internal and external data sources, analyzes the data, and displays it visually in an easy-to-understand format. Dashboards are usually customized to meet the specific needs of a business process, department, or the entire company.

There are many different types of analytics, including predictive, prescriptive, descriptive, and cognitive. Some analytics are named after their usage such as retail, supply chain, store optimization, sales force optimization, marketing optimization, call center optimization, web, social media, speech, credit risk, and fraud analytics. Analytics are also categorized by their use and characteristics such as actionable, visual, embedded, automated, and operational analytics.

Analytics can help improve decision making in many ways. At the most basic level, analytics can identify a problem or issue for management to resolve. At an intermediate level, analytics can also collect the data needed to solve the problem, analyze it, and make recommendations to management on how to resolve it. At an advanced level, actionable insights can be integrated into the systems used to make decisions. That is, the analytics can be embedded into AIS components, such as databases, applications, and devices, and operationalized to automatically resolve problems that occur and communicate solutions to management.

The data analytics market is estimated to exceed \$300 billion and is growing rapidly. Businesses use analytics to increase sales, create products and services to meet new customer needs, reduce costs, and improve decision making. Here are just a few ways data analytics are used:

- Cargill has developed an analytics platform that allows dairy farmers to use a tablet or computer to analyze large quantities of data about their cows' living conditions, diet, and milk productivity. When cows are happier and more comfortable, they produce more milk.
- Under Armour uses data from MapMyFitness to determine popular running routes and times so it knows when to advertise sneakers, energy drinks, and other products. Research shows that when an advertiser sells repeat-purchase products, ads sent at the right time to consumers will generate up to 16 times more sales than when sent at other times.
- Most web-based retailers such as Amazon use their sales database to suggest additional products and services for its customers to purchase.
- A technician at a support center uses key words from a caller, as well as a database of past problems and solutions, to quickly solve technical issues. In time, when the data and analytics are robust enough, the process will be automated; customers will visit the company's website and diagnose most problems themselves, with the most complex issues handed off to a human technician.
- An airline embeds sensors in all important airplane components, continuously monitors them, automatically schedules preventive maintenance, and provides a detailed list of what needs to be done.

Focus 1-2 explains how Walmart uses data analytics to improve their business.

THE AIS AND BLOCKCHAIN

In 2008, Satoshi Nakamoto invented the blockchain to digitally record cryptocurrency transactions such as Bitcoin. Since then, blockchain technology has been adapted so that virtually everything of value can be recorded, and private blockchains have been developed for business use.

Just as people do not have to know exactly how the Internet or an automobile works, they do not need to know the technical details of how the blockchain works to use it. However, a basic knowledge of blockchain technology helps users make better use of the technology.

Blockchain got its name from its structure, which is individual digital records, called *blocks*, linked together using cryptography in a single list, called a *chain*. The blockchain isn't

data dashboard - Interactive real-time display of key indicators of operating performance; display of important data points, metrics, and key performance indicators in easily understood data visualizations such as line or bar charts, tables, or gauges.

blockchain - Individual digital records, called *blocks*, linked together using cryptography in a single list, called a *chain*. The blockchain isn't stored in a single location. Instead, it is a distributed ledger of hashed documents that functions as a decentralized database. Each computer in the distributed peer-to-peer network maintains a copy of the ledger to prevent a single point of failure.

FOCUS 1-2 Data Analytics at Walmart

Walmart, the world's biggest retailer, has more than 245 million customers who shop at more than 20,000 stores in 28 countries. Walmart tracks individual customers, gathering data on what customers like and what they buy. Every hour, more than 1 million customers purchase something, which generates 2.5 petabytes (a quadrillion bytes) of unstructured data. The 2.5 petabytes are roughly 167 times the books contained in the U.S. Library of Congress.

To make use of this data, Walmart spent considerable resources to become a leader in big data analytics. In fact, Walmart's data analytics efforts started before the term *big data* became popular. Walmart's analytics systems analyze millions of products and hundreds of millions of customers using data gathered from many different internal and external data sources.

Data analytics support several Walmart goals. First, they optimize and personalize customers' shopping experiences, whether they are in a store, shopping online via a computer or a mobile device, or browsing Walmart's website. They analyze customer data and buying behaviors to better anticipate customer needs. Walmart also uses analytics to improve the store checkout experience.

Second, data analytics optimize operational efficiency, including increasing sales and profits, facilitating better and quicker decision making, improving store and employee efficiency, improving product assortment, managing the supply chain, and locating distribution centers and stores. Analytics helped Walmart (1) develop smarter stocking, pricing, merchandising, and marketing solutions in real time and (2) make its pharmacies more efficient.

Walmart analyzed sales before and after big data analytics were used to change their e-commerce strategy and found that online sales increased 10% to 15%, resulting in \$1 billion in incremental revenue.

The data analytics team at Walmart Labs monitors Walmart.com and analyzes every clickable action to

determine what consumers buy online and how to improve their online experience. The Lab also monitors what is trending on Twitter, local events such as sports activities and concerts that affect sales, and local weather deviations to determine how they affect buying patterns. Walmart's predictive data analytics software contains machine learning technologies that continuously improve the accuracy of analytics algorithms.

Walmart built the world's biggest private cloud to facilitate its data analytics processes. At its headquarters, Walmart also created a Data Café (Collaborative Analytics Facilities for Enterprise) to model, manipulate, and visualize data to create solutions. The Café uses Walmart's transaction data as well as data from 200 other data sources such as economic, gasoline, local events, social media, telecommunications, television, and weather data to predict outcomes and resolve issues to make Walmart stores more efficient, responsive, and profitable.

Employees are encouraged to submit their problems to Data Café experts for a solution usually produced in minutes rather than weeks and displayed on the Café's touchscreen smartboards. These real-time solutions help correct errors instantly, gain sales and marketing insights, and track customer trends and competitor strategies.

Here are two examples of Walmart's use of the Data Café. A grocery team could not figure out why sales in a product category had suddenly declined and used the Café to drill down into the data. They found that a pricing miscalculation resulted in the product being priced higher than it should have been in some stores. In a second example, Walmart data mining algorithms found that a specific cookie was popular at all Walmart stores except one. An investigation showed that a simple stocking oversight resulted in the cookies not being stocked on the store shelves. Ordering and stocking the cookies prevented the further loss of sales.

stored in a single location; it is a distributed ledger that functions as a decentralized database. Each computer in the distributed peer-to-peer network maintains a copy of the ledger to prevent a single point of failure. Since the blockchain is managed by a network that follows protocols for inter-node communication and validating new blocks, there is no need for a central authority that controls everything. The information recorded in the blockchain is made public, so everyone on the blockchain's peer-to-peer network has a copy of the blockchain and all transactions are accessible to everyone.

Here is a brief, high-level view of how the blockchain works; that is, how a transaction is added to a block and how a block is added to a chain. Blockchains are explained in more detail in **Chapter 12**.

1. **Initiate transaction.** Two parties, such as a buyer and a seller, decide to exchange something of value and request that a transaction be initiated. Instead of using actual buyer or

seller names, a unique digital signature or identifier is used. This is analogous to using a part number for a product or a username for a person.

2. **Validate transaction.** The transaction is sent to the peer-to-peer network nodes that use algorithms to simultaneously validate transaction details, including its time, dollar amount, and participants. To achieve a consensus, a simple majority of 51% of the nodes must validate the block. The number of computers in the peer-to-peer networks can be as large as desired; the Bitcoin blockchain has millions, each with a copy of its blockchain ledger. In a public, permission-less blockchain platform like Bitcoin, every network node can record transactions and participate in the consensus process. In a private, permissioned chain, participation in the consensus process is restricted to approved nodes.
3. **Create a block.** Since each block in a chain can store up to 1 MB of data, the verified transactions are combined with hundreds or thousands of similar transactions to create a new block for the ledger. The transaction's dollar amount and the digital signatures of both parties are stored in the block.
4. **Calculate and insert a hash.** Each block is given two unique codes, or pieces, of identifying information called a *hash*, which distinguishes it from other blocks. Hash codes use a mathematical algorithm to turn digital information into a string of numbers and letters. One hash is that of the current block, and the second is the hash of the block that precedes it in the chain. When a new block is added to the chain, it is linked to the previous block by storing a cryptographic hash generated from the contents of the previous block. The second hash ensures that the chain is never broken and that each block is recorded in a permanent and unalterable way. The second hash makes the block tamper resistant and secure; that is, it adds a high level of assurance that the prior block contents have not been changed. If the data on a block is edited in any way, that block's hash code changes, and the codes of all subsequent blocks change. This discrepancy makes it extremely difficult for information on the blockchain to be changed without notice. How a hash is created is discussed in **Chapter 12**.
5. **Complete transaction.** The block is added to the blockchain, and all the other computers storing the blockchain are updated automatically. This completes the transaction recording process, and the right of ownership of the item of value is passed from the seller to the buyer.

Blockchain has several significant advantages, including the following:

- **Accuracy.** Transactions are verified by many thousands of networked computers instead of error-prone humans. Even if a computer makes a computational mistake, the error would not spread to the rest of the blockchain unless at least 51% of the network's computers validated the mistake.
- **Transparency.** Blockchain data are transparent. That is, all transaction details, including participant user names, transaction amount, transaction date and time, and who entered the transaction, are open for everyone on the blockchain to see. This includes authorized regulators, auditors, etc.
- **Data consistency.** In older legacy systems, data are often located in multiple databases and finding data can be complex. The data can also be inconsistent among databases, with some of them updated and others not. With blockchain, data are stored in one location only.
- **Trust.** To ensure that blockchain networks can be trusted, computers that want to join the blockchain are tested. That is, new users are required to prove themselves before they can be a part of a blockchain network. For example, in Bitcoin's proof-of-work test, a system must expend significant computer power and energy to solve a complex math problem before a block can be added to the blockchain. While Bitcoin's proof of work does not make a hack impossible, the cost of organizing an attack would almost always outweigh the benefits that could be achieved from the attack.
- **No need for third parties.** The consensus process of all nodes in the network agreeing on the blockchain's content allows mutually distrustful parties to enter into transactions safely without trusted third parties.
- **Single set of books.** As both sides of a transaction are stored in a single source, that eliminates some of the need for a set of books for the buyer and for the seller. One set of books provides a trust level not present in current legacy systems.

- **Cost.** Blockchain eliminates the costs of human third-party verification and many transaction processing costs.
- **Decentralization.** By storing the blockchain on all network computers, the risks of data held centrally are eliminated. For example, if a copy of the blockchain is hacked or compromised, only that copy is affected. This reduces or eliminates the traditional requirement for file and database backups.
- **Efficiency.** Blockchain works all day, every day, and transactions can be finalized within minutes and considered secure in no more than a few hours. Contrast that with limited business hours and waiting days for transactions to clear and for money to be available.
- **Privacy.** Although many blockchains are public databases, where users can view transaction information, users are unable to access confidential data that identify those engaging in the transactions.
- **Security.** A blockchain is difficult to corrupt. There is no single point of failure; if one node goes down, there is a copy of the ledger on all the other nodes. Information is shared and continually reconciled by thousands of computers. New blocks are always added chronologically to the end of the blockchain. It is very difficult to go back and change a block's contents because each block contains its own hash and the hash of the previous block. If information is changed, the hashes for the previous and subsequent blocks also change, and this disrupts the ledger's shared state. When other network computers become aware that the change has caused a problem, consensus is no longer possible. Until the problem is solved, no new blocks are added to the blockchain. In most cases, the block that caused the error is discarded, and the nodes again attempt to achieve consensus. This process ensures that no single system or user can tamper with the transaction records or add invalid blocks to the blockchain.
- **Provenance.** Provenance is the history of ownership of something of value. The data collected by blockchain shows who did what, when they did it, and the history of the item since it was entered in the blockchain. That history is transparent, verified by all network participants, and frequently reconciled.

While blockchain has many important advantages, there are significant challenges to its adoption, including political and regulatory issues. These challenges include the following:

- **Cost.** Cost is a blockchain advantage because transaction fees are less, but it is also a disadvantage because the technology needed to operate a blockchain is expensive. So are the thousands of hours expended to produce the custom software and back-end processes needed to insert blockchain technologies into current business systems. There are also the utility costs required to run the computers that process and store the blockchain.
- **Loss of privacy and confidentiality.** Since blockchain users are unable to identify those engaging in transactions, dishonest users can use blockchain networks like Bitcoin to make illegal purchases. To prevent this, some countries like the United States prohibit full anonymity by requiring online exchanges to collect customer information, verify their identity, and confirm that they are not on any list of known or suspected terrorists.
- **Susceptibility.** A 51% attack is difficult to execute due to the computational power required to gain majority control of a blockchain network. That might change as technology costs decrease and hackers are able to affordably rent computational power rather than buying it. In addition, these attacks are more difficult as the number of nodes in the blockchain network increases.

In summary, a blockchain is a public, global, cryptographically secure ledger that automatically records and verifies large volumes of digital transactions. The combination of blocks that cannot be changed, blocks linked together in a chain, and cryptography to secure everything creates a transaction recording system that can be trusted for transactions among untrusted partners.

Focus 1-3 briefly discusses some of the many blockchain applications in use today.

FOCUS 1-3 Current and Planned Uses of Blockchain

Blockchain was first used for digital currencies but has spread to many industries because it is a secure and cost-effective way to manage digital transactions. Recent start-ups developed their business processes using newer technologies, so they are much more likely to use blockchain. As time passes, some of these blockchain-inspired start-ups become what some people call *emerging disruptors* as they begin displacing older and larger companies.

Established companies with older legacy information systems find it more challenging to use blockchain. They often discover that to achieve blockchain's advantages, they must change the way they do business.

Here are just a few of the more prominent uses of blockchain:

- Banks are only open during business hours and deposits take up to 3 days to clear. Blockchain is always open, and deposits can be seen in 10 minutes. It is estimated that blockchain could save banks \$20 billion by eliminating money in-transit costs and consumers \$16 billion in banking and insurance fees.
- Currently, healthcare data is stored by different institutions using different formats and standards, making it hard for doctors to understand a patient's medical history. Using blockchain technology, patient information can be signed, time-stamped, and stored securely on a distributed ledger. Doctors and patients can go to a

single source to access a patient's health information, and patients have more control over their own health data. Blockchain could also help solve the problem of counterfeit drugs in the medical supply chain.

- Recording property rights occurs when a deed is delivered to the local recording office and manually entered in the county's database and public index. The process is costly, inefficient, time-consuming, error prone, and susceptible to fraud. Disputed property claims are difficult to reconcile. Blockchain can eliminate scanning documents and finding paper files in a recording office. Property owners can trust that their deed is accurate and permanent if its ownership is stored and verified on the blockchain.
- In 2016, Overstock.com was one of the first publicly traded companies to use blockchain to sell and distribute company shares. Without blockchain, selling stock involves brokers, clearinghouses, and custodians. The money and shares involved in the trade are frozen for up to 3 days. With blockchain peer-to-peer trading, there is no need for intermediaries, and the shares exchange hands within minutes. Companies are also developing a blockchain application for proxy voting.
- Manufacturing companies are developing applications to track the flow of materials, information, and payments as they move through their supply chains.

CLOUD COMPUTING, VIRTUALIZATION, AND THE INTERNET OF THINGS

Recently, many organizations have embraced virtualization, cloud computing, and the Internet of Things to enhance both efficiency and effectiveness. **Virtualization** takes advantage of the power and speed of modern computers to run multiple systems simultaneously on one physical computer. This cuts hardware costs because fewer computers need to be purchased. Fewer machines means lower maintenance costs. Data center costs also fall because less space needs to be rented, which also reduces utility costs.

Cloud computing takes advantage of the high bandwidth of the modern global telecommunications network to enable employees to use a browser to remotely access software (software as a service), hardware (infrastructure as a service), and entire application environments (platform as a service). The arrangement is referred to as a "private," "public," or "hybrid" cloud, depending on whether the remotely accessed resources are entirely owned by the organization, a third party, or a mix of the two. Table 1-4 compares the different levels of service provided in the cloud to eating pizza. You can either make and bake a pizza at home, buy a frozen pizza and bake it, have a pizza delivered to your home, or go out to eat a pizza. As the table shows, you can do the same with computer services (the items in blue are done at home, and the items in red are done by a cloud provider).

Cloud computing can potentially generate significant cost savings. For example, instead of purchasing, installing, and maintaining separate copies of software for each end user, an organization can purchase one copy, install it on a central server, and pay for the right for a specified number of employees to simultaneously use a browser to remotely access and use that software.

virtualization - Running multiple systems simultaneously on one physical computer.

cloud computing - Using a browser to remotely access software, data storage, hardware, and applications.

TABLE 1-4 Comparing Cloud Services to Eating Pizza

Traditional (make pizza from scratch at home)	IaaS: Infrastructure as a Service (take and bake)	PaaS: Platform as a Service (pizza delivered)	SaaS: Software as a Service (dine out)
Dining table			
Soda			
Electricity and gas			
Oven			
Dough			
Toppings			

Public clouds eliminate the need for making major capital investments in IT, with organizations purchasing (and expensing) their use of computing resources on a pay-for-use or subscription basis. In addition to reducing costs, the centralization of computing resources with cloud computing (whether public, private, or hybrid) makes it easier to change software and hardware, thereby improving flexibility. The term **Internet of Things (IoT)** refers to the embedding of sensors in a multitude of devices (lights, heating and air conditioning, appliances, etc.) so that those devices can now connect to the Internet. The IoT has significant implications for information security.

Internet of Things (IoT) - Embedding sensors in devices so they can connect to the Internet.

THE AIS AND CORPORATE STRATEGY

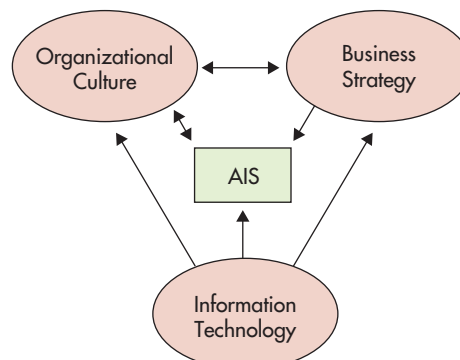
Since most organizations have limited resources, it is important to identify the AIS improvements likely to yield the greatest return. Making a wise decision requires strategic thinking, planning, and an understanding of the organization's overall business strategy.

Figure 1-5 shows three factors that influence the design of an AIS: developments in IT, business strategy, and organizational culture. It is also important to recognize that the design of the AIS can also influence the organization's culture by controlling the flow of information within the organization. For example, an AIS that makes information easily accessible and widely available is likely to increase pressures for more decentralization and autonomy.

IT developments affect business strategy. For example, the Internet has affected the way many activities are performed, significantly influencing both strategy and strategic positioning. The Internet dramatically cuts costs, thereby helping companies to implement a low-cost strategy. If every company used the Internet to adopt a low-cost strategy, then the effects might be problematic. Indeed, one possible outcome may be intense price competition among firms, with the likely result that most of the cost savings provided by the Internet get passed on to the industry's customers, rather than being retained in the form of higher profits. Moreover, because every company can use the Internet to streamline its activities, a company is unlikely to gain a sustainable long-term competitive advantage.

An organization's AIS plays an important role in helping it adopt and maintain a strategic position. Achieving a close fit among activities requires that data be collected about each

FIGURE 1-5
Factors Influencing
Design of the AIS



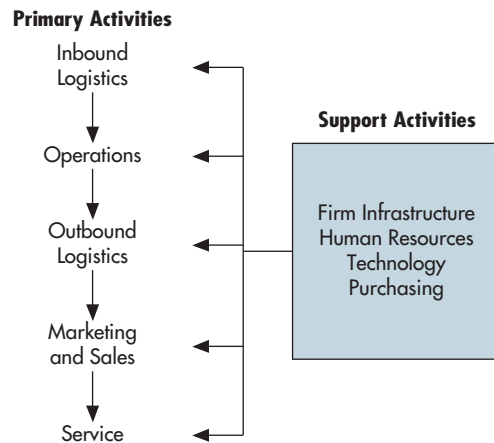


FIGURE 1-6
The Value Chain

activity. It is also important that the information system collect and integrate both financial and nonfinancial data about the organization's activities.

THE ROLE OF THE AIS IN THE VALUE CHAIN

To provide value to customers, most organizations perform a number of different activities. Figure 1-6 shows these activities can be conceptualized as forming a **value chain** consisting of five **primary activities** that directly provide value to customers:

1. **Inbound logistics** consists of receiving, storing, and distributing the materials an organization uses to create the services and products it sells. For example, an automobile manufacturer receives, handles, and stores steel, glass, and rubber.
2. **Operations** activities transform inputs into final products or services. For example, assembly line activities convert raw materials into a finished car and retailers remove goods from packing boxes and place the individual items on shelves for customers to purchase.
3. **Outbound logistics** activities distribute finished products or services to customers. An example is shipping automobiles to car dealers.
4. **Marketing and sales** activities help customers buy the organization's products or services. Advertising is an example of a marketing and sales activity.
5. **Service** activities provide post-sale support to customers. Examples include repair and maintenance services.

Support activities allow the five primary activities to be performed efficiently and effectively. They are grouped into four categories:

1. **Firm infrastructure** is the accounting, finance, legal, and general administration activities that allow an organization to function. The AIS is part of the firm infrastructure.
2. **Human resources** activities include recruiting, hiring, training, and compensating employees.
3. **Technology** activities improve a product or service. Examples include research and development, investments in IT, and product design.
4. **Purchasing** activities procure raw materials, supplies, machinery, and the buildings used to carry out the primary activities.

Using IT to redesign supply chain systems yields tremendous benefits and cost savings. For example, Tennessee Valley Authority, a power generator, reengineered its supply chain and created an enterprise-wide system that provides up-to-the-minute information rather than the "current once a day" system that it replaced. The new system replaced 20 smaller and incompatible systems, reduced head count by 89 people, and saved \$270 million in its first five years.

An organization's value chain is a part of a larger system called a **supply chain**. As shown in Figure 1-7, a manufacturing organization interacts with its suppliers and distributors. By paying attention to its supply chain, a company can improve its performance by helping the

value chain - Linking all primary and support activities in a business. Value is added as a product passes through the chain.

primary activities - Value chain activities that produce, market, and deliver products and services to customers and provide postdelivery service and support.

support activities - Value chain activities such as firm infrastructure, technology, purchasing, and human resources that enable primary activities to be performed efficiently and effectively.

supply chain - An extended system that includes an organization's value chain as well as its suppliers, distributors, and customers.

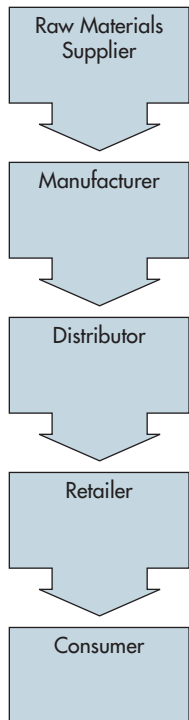


FIGURE 1-7
The Supply Chain

others in the supply chain to improve their performance. For example, S&S can improve its purchasing and inbound logistics activities by implementing a more efficient just-in-time inventory management system that reduces its costs and minimizes the capital tied up in inventory. S&S can reap additional benefits if it links its new systems with its suppliers so they can perform their primary value chain activities more efficiently. For example, by providing more detailed and timely information about its inventory needs, S&S suppliers can more efficiently plan their production schedules. Part of the resultant cost reduction can be passed on to S&S in the form of lower product costs.

The problems created by an ineffective supply chain are illustrated by Limited Brands. A number of acquisitions left Limited with a tangled web of more than 60 incompatible information systems. The problems came to a head one night when 400 trailers converged on a distribution center parking lot that could fit only 150 trailers. The trailers blocked traffic along all the highways around the distribution center and caused countless traffic and community problems. No one from Limited knew where all the trailers came from, what the merchandise was, or where it was to be sent. Chaos reigned until the merchandise could be routed to stores and other distribution centers. Limited solved many of its problems by installing a new, integrated system that greatly improved its supply chain processes and technologies. Developing the new system was not easy. Limited had more than 1,000 suppliers and sells its merchandise using various platforms, including retail stores, the Internet, catalogs, and third-party retailers. Limited subsequently sold many of its acquisitions and re-branded itself as L Brands.

Summary and Case Conclusion

Silvia and Sundip reflected on what they had done to try and understand what decisions S&S would need to make and the information needed to make them. They began by obtaining an understanding of S&S's basic business processes and key decisions that must be made to operate the business effectively. They followed that with an analysis of the internal and external parties that the AIS would have to interact with and the information the AIS would have to provide them.

Since S&S is a retail merchandising company, its business processes could be described in terms of four basic transaction cycles:

1. The *revenue cycle* encompasses all transactions involving sales to customers and the collection of cash receipts for those sales.
2. The *expenditure cycle* encompasses all transactions involving the purchase and payment of merchandise sold by S&S as well as other services it consumes such as rent and utilities.
3. The *human resources/payroll cycle* encompasses all the transactions involving the hiring, training, and payment of employees.
4. The *financing cycle* encompasses all transactions involving the investment of capital in the company, borrowing money, payment of interest, and loan repayments.

These four cycles interface with the *general ledger and reporting system*, which consists of all activities related to the preparation of financial statements and other managerial reports.

Sundip and Silvia will need a well-designed AIS to provide the information necessary to effectively plan, manage, and control their business. Their AIS must be able to process data about sales and cash receipts, purchasing and paying for merchandise and services, payroll and tax-related transactions, and acquiring and paying for fixed assets. The company's AIS must also provide the information needed to prepare financial statements.

Fortunately, many computer-based accounting packages are available for the retail industry. As they begin looking at various software packages, however, they quickly learn that considerable accounting knowledge is required to choose the one that will best fit their business. Because neither has an accounting background, they decide their next task is to hire an accountant.

KEY TERMS

system 3	give-get exchange 7	accounting 10
goal conflict 3	business processes or transaction cycles 7	artificial intelligence (AI) 12
goal congruence 3	revenue cycle 7	data analytics 12
data 3	expenditure cycle 7	data dashboard 13
information 3	production cycle 8	blockchain 13
machine-readable 4	human resources/payroll cycle 8	virtualization 17
information overload 4	financing cycle 8	cloud computing 17
information technology (IT) 4	general ledger and reporting system 9	Internet of Things (IoT) 18
value of information 4	accounting information system (AIS) 10	value chain 19
information system 5		primary activities 19
business process 5		support activities 19
transaction 7		supply chain 19
transaction processing 7		

AIS in Action**CHAPTER QUIZ**

- Data differ from information in which way?
 - Data are output, and information is input.
 - Information is output, and data are input.
 - Data are meaningful bits of information.
 - There is no difference.
- Which of the following is NOT a characteristic that makes information useful?
 - It is accurate.
 - It is timely.
 - It is inexpensive.
 - It is relevant.
- Which of the following is a primary activity in the value chain?
 - purchasing
 - accounting
 - post-sales service
 - human resource management
- Which transaction cycle includes interactions between an organization and its suppliers?
 - revenue cycle
 - expenditure cycle
 - human resources/payroll cycle
 - general ledger and reporting system
- Which of the following is NOT a means by which information improves decision making?
 - increases information overload
 - reduces uncertainty
 - provides feedback about the effectiveness of prior decisions
 - identifies situations requiring management action
- In the value chain concept, upgrading IT is considered what kind of activity?
 - primary activity
 - support activity
 - service activity
 - structured activity
- In which cycle does a company ship goods to customers?
 - production cycle
 - financing cycle
 - revenue cycle
 - expenditure cycle
- Which of the following is a function of an AIS?
 - reducing the need to identify a strategy and strategic position
 - transforming data into useful information
 - allocating organizational resources
 - automating all decision making

9. A firm, its suppliers, and its customers collectively form which of the following?
 - a. supply chain
 - b. value chain
 - c. ERP system
 - d. AIS
10. A performance report about all approved vendors during the previous 12 months is information MOST needed in which business process?
 - a. paying vendors
 - b. acquiring inventory
 - c. selling merchandise
 - d. paying employees

DISCUSSION QUESTIONS

- 1.1 The value of information is the difference between the benefits realized from using that information and the costs of producing it. Would you, or any organization, ever produce information if its expected costs exceeded its benefits? If so, provide some examples. If not, why?
- 1.2 Can the characteristics of useful information listed in Table 1-1 be met simultaneously? Or does achieving one mean sacrificing another?
- 1.3 You and a few of your classmates decided to become entrepreneurs. You came up with a great idea for a new mobile phone application that you think will make lots of money. Your business plan won second place in a local competition, and you are using the \$10,000 prize to support yourselves as you start your company.
 - a. Identify the key decisions you need to make to be successful entrepreneurs, the information you need to make them, and the business processes you will need to engage in.
 - b. Your company will need to exchange information with various external parties. Identify the external parties, and specify the information received from and sent to each of them.
- 1.4 How do an organization's business processes and lines of business affect the design of its AIS? Give several examples of how differences among organizations are reflected in their AIS.
- 1.5 Figure 1-5 shows that organizational culture and the design of an AIS influence one another. What does this imply about the degree to which an innovative system developed by one company can be transferred to another company?
- 1.6 Figure 1-5 shows that developments in IT affect both an organization's strategy and the design of its AIS. How can a company determine whether it is spending too much, too little, or just enough on IT?
- 1.7 Apply the value chain concept to S&S. Explain how it would perform the various primary and support activities.
- 1.8 IT enables organizations to easily collect large amounts of information about employees. Discuss the following issues:
 - a. To what extent should management monitor employees' e-mail?
 - b. To what extent should management monitor which websites employees visit?
 - c. To what extent should management monitor employee performance by, for example, using software to track keystrokes per hour or some other unit of time? If such information is collected, how should it be used?
 - d. Should companies use software to electronically "shred" all traces of e-mail?
 - e. Under what circumstances and to whom is it appropriate for a company to distribute information it collects about the people who visit its website?

PROBLEMS

- 1.1** IT is continually changing the nature of accounting and the role of accountants. Write a two-page report describing what you think the nature of the accounting function and the AIS in a large company will be like in the year 2030.
- 1.2** The annual report is considered by some to be the single most important printed document that companies produce. In recent years, annual reports have become large documents. They now include such sections as letters to the stockholders, descriptions of the business, operating highlights, financial review, management discussion and analysis, a discussion of company internal controls, segment reporting, inflation data, and the basic financial statements. The expansion has been due in part to a general increase in the degree of sophistication and complexity in accounting standards and disclosure requirements for financial reporting.

The expansion also is reflective of the change in the composition and level of sophistication of users. Current users include not only stockholders but also financial and securities analysts, potential investors, lending institutions, stockbrokers, customers, employees, and—whether the reporting company likes it or not—competitors. Thus, a report originally designed as a device for communicating basic financial information now attempts to meet the diverse needs of an ever-expanding audience.

Users hold conflicting views on the value of annual reports. Some argue that they fail to provide enough information, whereas others believe disclosures in annual reports have expanded to the point where they create information overload. Others argue that the future of most companies depends on acceptance by the investing public and by its customers; therefore, companies should take this opportunity to communicate well-defined corporate strategies.

REQUIRED

- Identify and discuss the basic factors of communication that must be considered in the presentation of the annual report.
 - Discuss the communication problems a corporation faces in preparing the annual report that result from the diversity of the users.
 - Select two types of information found in an annual report, other than the financial statements and accompanying footnotes, and describe how they are helpful to the users of annual reports.
 - Discuss at least two advantages and two disadvantages of stating well-defined corporate strategies in the annual report.
 - Evaluate the effectiveness of annual reports in fulfilling the information needs of the following current and potential users: shareholders, creditors, employees, customers, and financial analysts.
 - Annual reports are public and accessible to anyone, including competitors. Discuss how this affects decisions about what information should be provided in annual reports. (CMA Examination, adapted)
- 1.3** United Services Automotive Association (USAA) is one of the largest diversified financial services companies in the United States, with close to \$111 billion in assets under management. One reason for its success is the use of IT to lower costs and improve customer service. USAA operates one of the most advanced and successful information systems in the world. Most of its communication with its widely scattered customers, mostly military officers and their families, is digital. It was the first U.S. bank to implement a remote deposit capture application for the iPhone.

Early on, USAA made a strategic choice to become one of the more technology-intensive companies in the world. It views IT as a strategic weapon and uses it in several ways, including the following:

- When customers call, USAA personnel greet them personally by name. Unlike many diversified companies, a customer representative can handle inquiries and transactions about all of USAA's products using a highly integrated database.
- USAA uses its extensive database to keep track of minute details, such as which auto parts are fixed most frequently. It also uses its database to find ways to reduce claims costs. For example, USAA discovered that repair shops would rather charge up to \$300 to replace a windshield with punctures than to charge \$40 to repair it. USAA began offering to waive the deductible if the owners would repair their windshield rather than replace it.
- USAA spent extensively to develop an image-processing system that digitizes all paper documents sent in by claimants. It takes only a few keystrokes for a policy service representative to retrieve pictures of all the documents in a customer's file. The system can sort and prioritize documents so that employees are always working on the most important and urgent tasks.
- USAA is a world leader in mobile banking. Customers can use their cell phones and other mobile devices to access and execute banking, investment, stock trading, and insurance applications such as filing claims. More than 70% of USAA's logins are from cell phone users.
- USAA was among the first financial institutions to use text and e-mail messaging and notification technologies, person-to-person payment applications, and social networking and personal financial management tools connected to bank accounts.

REQUIRED

- a. Why should USAA collect data on which auto parts are fixed most frequently? What could it do with this data?
- b. Even though USAA offered to waive the deductible, the repair shops still managed to convince 95% of the owners to replace rather than repair their damaged windshields. How could USAA use its AIS to persuade more shop owners to repair rather than replace their windshields?
- c. How does the image-processing system at USAA add value to the organization?
- d. How do the remote deposit capture and mobile banking system at USAA add value to the organization?
- e. Do an Internet search and find out what other advancements USAA has introduced. For example, see if you can find out how USAA is using artificial intelligence, data analytics, blockchain, or some other emerging technology. Write a one-page report on each new application or other newsworthy item you find (maximum limit of three applications or items).

- 1.4.** Match the description listed in the right column with the correct information characteristic listed in the left column.

_____ 1. access restricted	a. A report was carefully designed so that its data was easily comprehended by the reader.
_____ 2. accurate	b. A manager working on the weekend needed information about a customer's production requests and found it on the company's network.
_____ 3. available	c. Before production reports are accepted, two clerks working independently must produce the same information.
_____ 4. complete	d. A report was checked by three different people to make sure it was correct.

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| _____ 5. concise | e. An accounts receivable aging report is used in a credit-granting decision. |
| _____ 6. consistent | f. An accounts receivable aging report was received before the credit manager had to decide whether to extend credit to a customer. |
| _____ 7. current | g. Needing help with a decision, a manager sought the opinion of a highly regarded expert. |
| _____ 8. objective | h. To protect intellectual property, a company encrypted the data, stored it in a very secure facility, and limited its use to five people. |
| _____ 9. relevant | i. Tired of keying supplier prices into a database, a purchasing manager insisted the data be sent in machine-readable form. |
| _____ 10. reputable | j. An accounts receivable aging report included all customer accounts. |
| _____ 11. timely | k. After a lengthy, rambling presentation, a CEO insisted future presentations contain only pertinent facts and last no more than 30 minutes. |
| _____ 12. useable | l. A new manager insisted that monthly reports look the same so she could compare a new month's results to those of previous months. |
| _____ 13. understandable | m. After making a decision based on outdated data, a new CEO required all analyses to be conducted with up-to-date data. |
| _____ 14. verifiable | n. Reluctant to rely on his personal feelings about a decision, a manager sought the opinion of an outside expert. |

- 1.5.** Classify each of the following items as belonging in the revenue, expenditure, human resources/payroll, production, or financing cycle.
- Collect payment on customer accounts.
 - Complete a picking ticket for a customer order.
 - Decide how many units to make next month.
 - Disburse payroll checks to factory workers.
 - Draw upon line of credit.
 - Establish a \$10,000 credit limit for a new customer.
 - Hire a new assistant controller.
 - Obtain a bank loan.
 - Pay federal payroll taxes.
 - Pay for raw materials.
 - Pay off mortgage on factory.
 - Pay property taxes on office building.
 - Pay sales commissions.
 - Pay utility bills.
 - Purchase raw materials.
 - Put purchased goods into the warehouse.
 - Record factory employee timecards.
 - Record goods received from vendor.
 - Sell concert tickets.
 - Sell a DVD player.
 - Send an order to a vendor.
 - Send new employees to a business ethics course.
 - Update the allowance for uncollectible accounts.

1.6. This chapter discusses several technologies that impact accounting information systems: XBRL, artificial intelligence, data analytics, and blockchain. Research and then write a three-page report on one of these four technologies. Be sure to describe the technology's current impact on the AIS and how it is used in actual businesses or how an attempt to implement it failed.

1.7. Match each of the following terms with its definition.

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| _____ 1. | accounting information system | a. Exceeding the amount of information a human mind can absorb and process |
| _____ 2. | artificial intelligence | b. Use of software and algorithms to find and solve problems and improve business performance |
| _____ 3. | blockchain | c. The benefit provided by information minus the cost of producing it |
| _____ 4. | business processes | d. A set of activities and tasks that help accomplish a specific organizational goal |
| _____ 5. | data | e. An agreement to exchange goods or services in exchange for cash |
| _____ 6. | data analytics | f. Process of capturing, processing, and storing transaction data for later use and for producing information output |
| _____ 7. | data dashboard | g. Frequent exchanges such as surrendering cash for inventory and paying employees for labor |
| _____ 8. | expenditure cycle | h. Activities associated with selling goods and services in exchange for cash or a future promise of cash |
| _____ 9. | general ledger and reporting system | i. Value chain activities that produce, market, and deliver products to customers and provide post-delivery support |
| _____ 10. | give-get exchange | j. A system that collects, records, stores, and processes data to produce information for decision makers |
| _____ 11. | goal congruence | k. Linking all the primary and support activities in a business |
| _____ 12. | information | l. Activities such as firm infrastructure and technology that enable main activities to be performed efficiently and effectively |
| _____ 13. | machine-readable | m. An organization's value chain as well as its vendors, distributors, and customers |
| _____ 14. | primary activities | n. Activities associated with purchasing inventory for resale for cash or a promise to pay cash |
| _____ 15. | production cycle | o. Data in a format that can be processed by a computer |
| _____ 16. | revenue cycle | p. Computer systems that simulate human intelligence processes such as learning, reasoning, and self-improvement |
| _____ 17. | supply chain | q. Procedures and routines that carry out specific activities, achieve objectives, or solve problems |
| _____ 18. | support activities | r. Facts collected, recorded, stored, and processed by an information system |
| _____ 19. | system | s. Activities associated with using labor, raw materials, and equipment to produce finished goods |
| _____ 20. | transaction | t. Organized and processed data that provide meaning and improve decision making |

- _____ 21. value chain
- _____ 22. value of information
 - u. Display of data points and performance indicators in easily understood charts, tables, or gauges
 - v. Activities associated with hiring, compensating, promoting, and terminating employees
 - w. Major give-get exchanges that occur frequently in most companies
 - x. Information-processing operations involved in preparing reports for internal and external parties
 - y. Individual digital records linked using cryptography in a single list called a *chain*
 - z. A subsystem achieves its goals while contributing to the overall goal

AIS in Action Solutions

QUIZ KEY

1. Data differ from information in which way?
 - a. Data are output, and information is input. [Incorrect. Data are facts and figures that, once organized, can become information. Therefore, data are inputs, and information is output.]
 - ▶ b. Information is output, and data are input. [Correct.]
 - c. Data are meaningful bits of information. [Incorrect. Information is organized and processed data that provide meaning.]
 - d. There is no difference. [Incorrect. There is a difference. Data are unorganized facts and figures. Information is meaningful, organized, and processed data.]
2. Which of the following is NOT a characteristic that makes information useful?
 - a. It is accurate. [Incorrect. This is one of the information characteristics listed in Table 1-1.]
 - b. It is timely. [Incorrect. This is one of the information characteristics listed in Table 1-1.]
 - ▶ c. It is inexpensive. [Correct. This is NOT one of the information characteristics listed in Table 1-1.]
 - d. It is relevant. [Incorrect. This is one of the information characteristics listed in Table 1-1.]
3. Which of the following is a primary activity in the value chain?
 - a. purchasing [Incorrect. This is a support activity.]
 - b. accounting [Incorrect. This is a firm infrastructure support activity.]
 - ▶ c. post-sales service [Correct. Service is a primary activity.]
 - d. human resource management [Incorrect. This is a support activity.]
4. Which transaction cycle includes interactions between an organization and its suppliers?
 - a. revenue cycle [Incorrect. The revenue cycle involves interactions between an organization and its customers.]
 - ▶ b. expenditure cycle [Correct.]
 - c. human resources/payroll cycle [Incorrect. The human resources/payroll cycle involves interactions between an organization and its employees, government, and potential hires.]
 - d. general ledger and reporting system [Incorrect. The general ledger and reporting system receives summary information from all cycles.]

5. Which of the following is NOT a means by which information improves decision making?
- ▶ a. increases information overload [Correct. Decision makers receiving too much information have difficulty incorporating all of the information into their decision framework, and, as a result, decision quality can be reduced rather than improved.]
 - b. reduces uncertainty [Incorrect. More reliable information leads to less uncertainty and thus better decisions.]
 - c. provides feedback about the effectiveness of prior decisions [Incorrect. Knowledge of effective and ineffective decisions can lead to better decisions in the future.]
 - d. identifies situations requiring management action [Incorrect. Identifying the need for management action can lead to improved decision making.]
6. In the value chain concept, upgrading IT is considered what kind of activity?
- a. primary activity [Incorrect. Investing in IT is a support activity.]
 - ▶ b. support activity [Correct. Technology activities, including investing in IT, are considered a support activity.]
 - c. service activity [Incorrect. The value chain includes only primary and support activities. A service activity is a type of primary activity.]
 - d. structured activity [Incorrect. The value chain includes only primary and support activities. A structured activity is neither a primary nor a secondary activity.]
7. In which cycle does a company ship goods to customers?
- a. production cycle [Incorrect. The production cycle involves the transformation of raw materials into finished goods.]
 - b. financing cycle [Incorrect. The financing cycle deals with interactions between an organization and its lenders and owners.]
 - ▶ c. revenue cycle [Correct. The revenue cycle involves interactions between an organization and its customers, such as shipping them goods.]
 - d. expenditure cycle [Incorrect. The expenditure cycle involves interactions between an organization and its suppliers.]
8. Which of the following is a function of an AIS?
- a. reducing the need to identify a strategy and strategic position [Incorrect. An AIS does not reduce the need to identify a strategy. It provides information to executives for the purpose of making strategic decisions.]
 - ▶ b. transforming data into useful information [Correct. This is one of the primary functions of an AIS.]
 - c. allocating organizational resources [Incorrect. Decision makers allocate resources, and the purpose of the AIS is to provide information to the decision makers so that they can make the allocation.]
 - d. automating all decision making [Incorrect. The AIS provides information to decision makers; it is not designed to automate all decision making.]
9. A firm, its suppliers, and its customers collectively form which of the following?
- ▶ a. supply chain [Correct. The supply chain is made up of the firm, its suppliers, and customers.]
 - b. value chain [Incorrect. The value chain is made up of primary and support activities within the firm.]
 - c. ERP system [Incorrect. An ERP system integrates all aspects of an organization's activities into one system.]
 - d. AIS [Incorrect. The AIS is made up of the human and capital resources within an organization that are responsible for collecting and processing transactions and preparing financial information.]

10. A performance report about all approved vendors during the previous 12 months is information MOST needed in which business process?
- a. paying vendors [Incorrect. To pay a vendor, a company needs to know whether merchandise ordered was received in good condition. They do not need a 12-month history of vendor performance.]
 - ▶ b. acquiring inventory [Correct. Companies want to acquire inventory from companies that have performed well in the past. A vendor performance report would disclose whether the vendor shipped inventory on time, whether the inventory was of the requested quality, whether the prices were as agreed upon, etc.]
 - c. selling merchandise [Incorrect. A 12-month history of vendor performance is usually not very helpful in trying to sell products to customers. More important would be customer tastes and preferences, customer credit status, etc.]
 - d. paying employees [Incorrect. It is very rare for an employee's pay to be based on a 12-month history of vendor performance. More important are hours worked, annual salary, sales figures to calculate commissions, etc.]